

# Multiplexer

A multiplexer is a combinational circuit that has  $2^n$  input lines and a single output line. Simply, the multiplexer is a multi-input and single-output combinational circuit. The binary information is received from the input lines and directed to the output line. On the basis of the values of the selection lines, one of these data inputs will be connected to the output.

Unlike encoder and decoder, there are  $n$  selection lines and  $2^n$  input lines. So, there is a total of  $2^N$  possible combinations of inputs. A multiplexer is also treated as **Mux**.

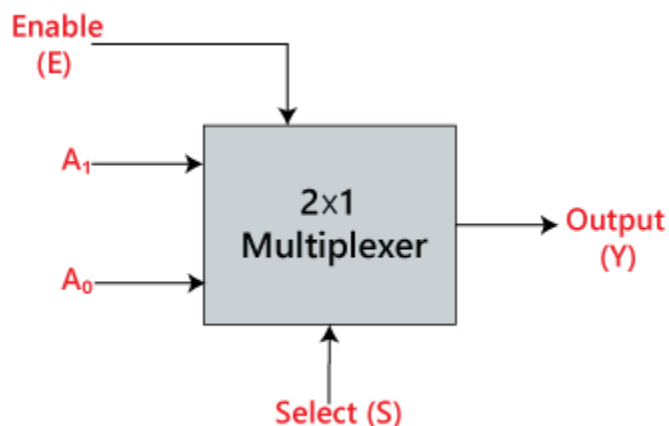
There are various types of the multiplexer which are as follows:

## 2×1 Multiplexer:

In 2×1 multiplexer, there are only two inputs, i.e.,  $A_0$  and  $A_1$ , 1 selection line, i.e.,  $S_0$  and single outputs, i.e.,  $Y$ . On the basis of the combination of inputs which are present at the selection line  $S^0$ , one of these 2 inputs will be connected to the output. The block diagram and the truth table of the 2×1 multiplexer are given below.

Backward Skip 10sPlay VideoForward Skip 10s

### Block Diagram:



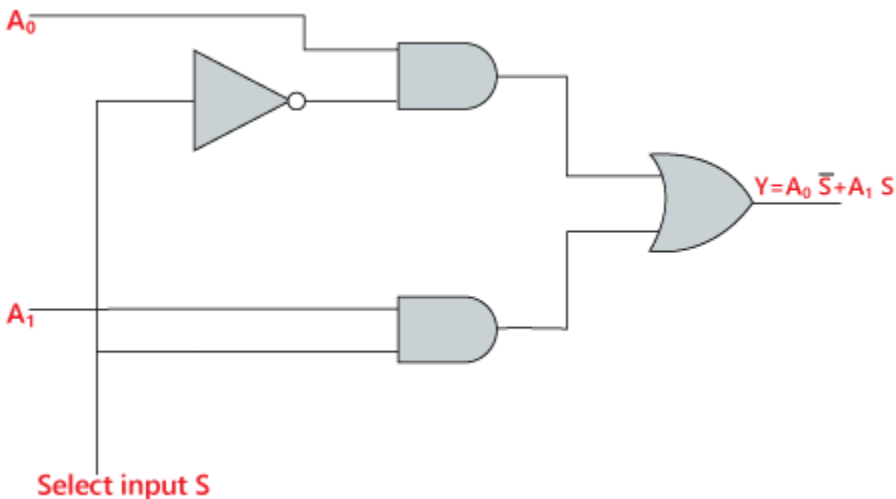
### Truth Table:

| INPUTS | Output |
|--------|--------|
| $S_0$  | Y      |
| 0      | $A_0$  |
| 1      | $A_1$  |

The logical expression of the term Y is as follows:

$$Y = S_0' \cdot A_0 + S_0 \cdot A_1$$

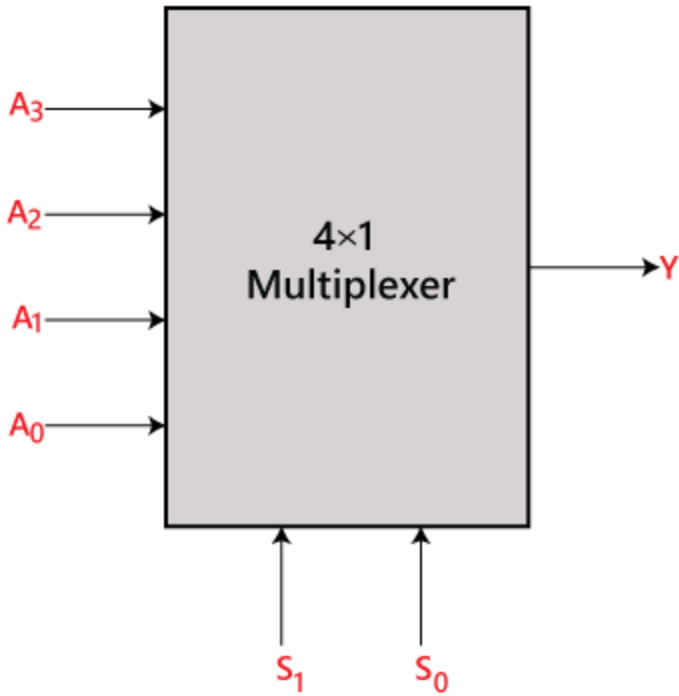
Logical circuit of the above expression is given below:



## 4×1 Multiplexer:

In the 4×1 multiplexer, there is a total of four inputs, i.e.,  $A_0$ ,  $A_1$ ,  $A_2$ , and  $A_3$ , 2 selection lines, i.e.,  $S_0$  and  $S_1$  and single output, i.e., Y. On the basis of the combination of inputs that are present at the selection lines  $S_0$  and  $S_1$ , one of these 4 inputs are connected to the output. The block diagram and the truth table of the 4×1 multiplexer are given below.

### Block Diagram:



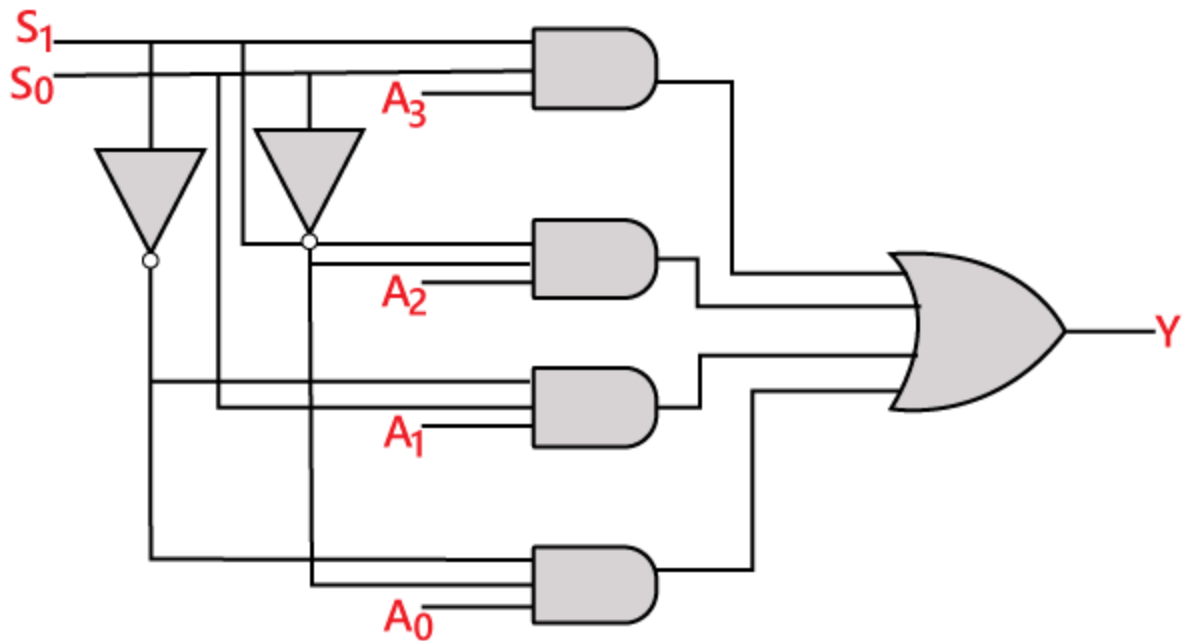
Truth Table:

| INPUTS |       | Output |
|--------|-------|--------|
| $S_1$  | $S_0$ | $Y$    |
| 0      | 0     | $A_0$  |
| 0      | 1     | $A_1$  |
| 1      | 0     | $A_2$  |
| 1      | 1     | $A_3$  |

The logical expression of the term  $Y$  is as follows:

$$Y = S_1' S_0' A_0 + S_1' S_0 A_1 + S_1 S_0' A_2 + S_1 S_0 A_3$$

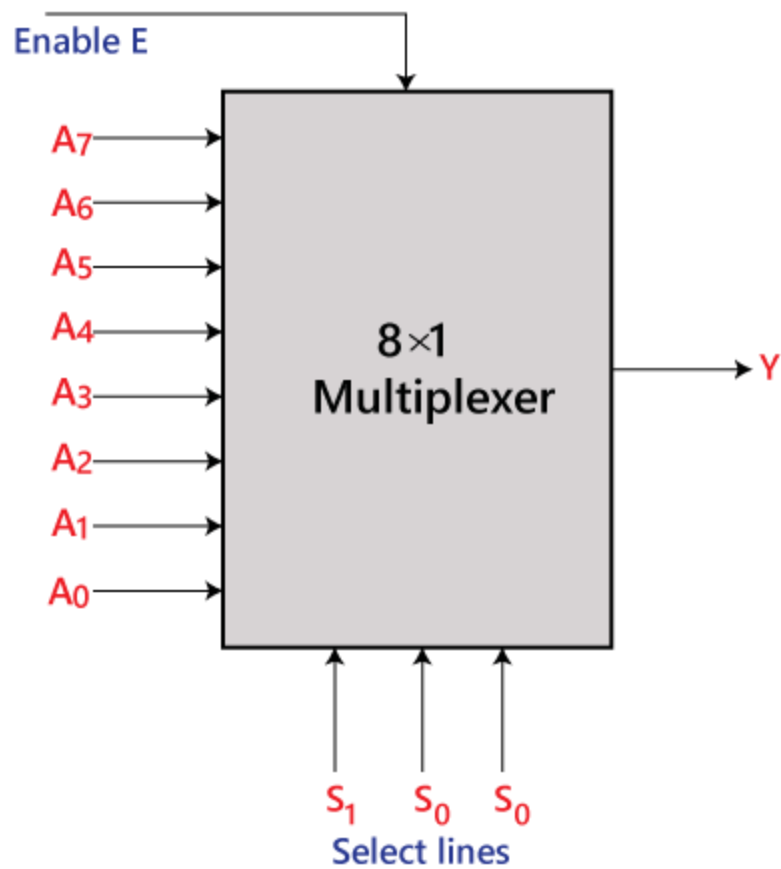
Logical circuit of the above expression is given below:



## 8 to 1 Multiplexer

In the 8 to 1 multiplexer, there are total eight inputs, i.e.,  $A_0, A_1, A_2, A_3, A_4, A_5, A_6$ , and  $A_7$ , 3 selection lines, i.e.,  $S_0, S_1$  and  $S_2$  and single output, i.e.,  $Y$ . On the basis of the combination of inputs that are present at the selection lines  $S^0, S^1$ , and  $S_2$ , one of these 8 inputs are connected to the output. The block diagram and the truth table of the  $8 \times 1$  multiplexer are given below.

**Block Diagram:**



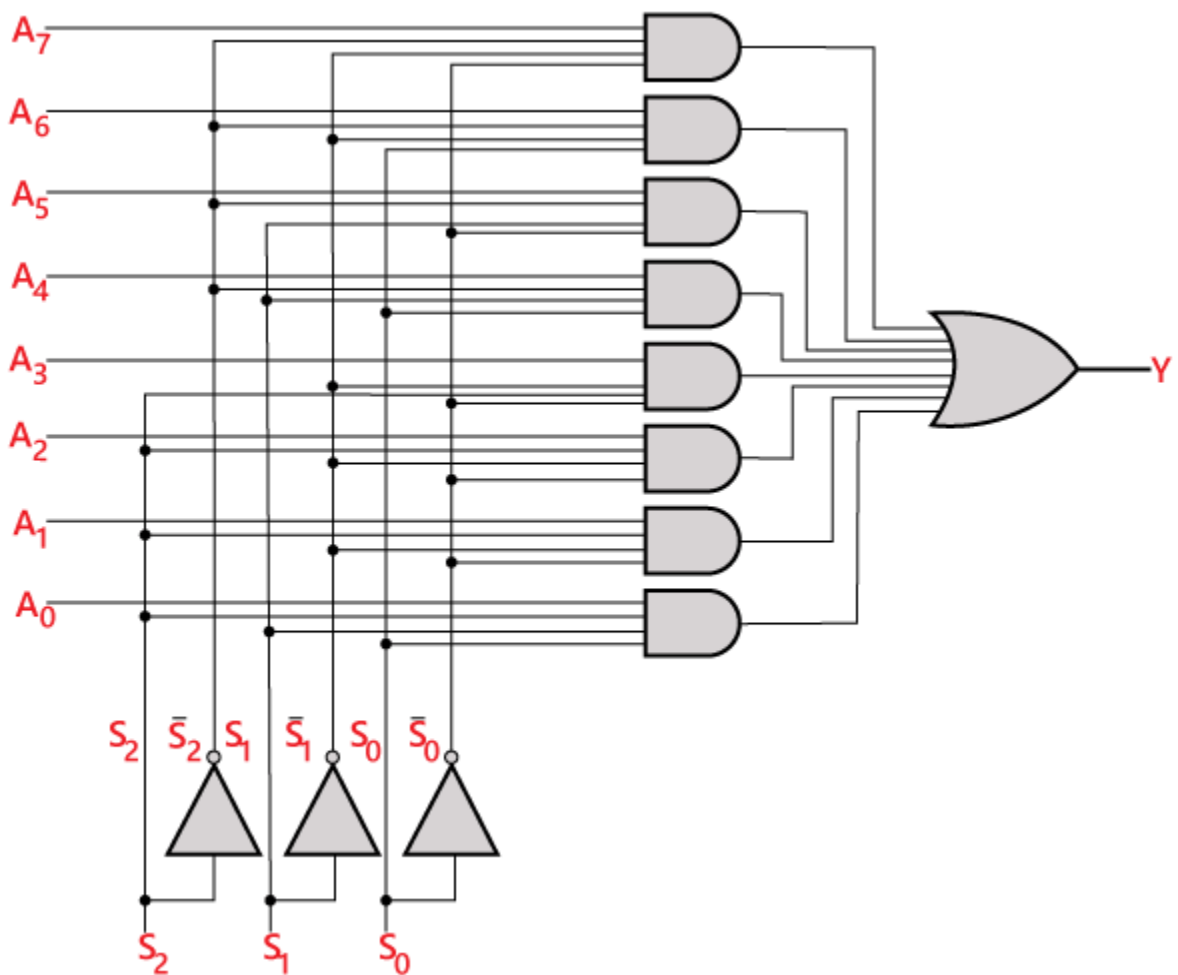
Truth Table:

| INPUTS |       |       | Output |
|--------|-------|-------|--------|
| $S_2$  | $S_1$ | $S_0$ | $Y$    |
| 0      | 0     | 0     | $A_0$  |
| 0      | 0     | 1     | $A_1$  |
| 0      | 1     | 0     | $A_2$  |
| 0      | 1     | 1     | $A_3$  |
| 1      | 0     | 0     | $A_4$  |
| 1      | 0     | 1     | $A_5$  |
| 1      | 1     | 0     | $A_6$  |
| 1      | 1     | 1     | $A_7$  |

The logical expression of the term Y is as follows:

$$Y = S_0' \cdot S_1' \cdot S_2' \cdot A_0 + S_0 \cdot S_1' \cdot S_2' \cdot A_1 + S_0' \cdot S_1 \cdot S_2' \cdot A_2 + S_0 \cdot S_1 \cdot S_2' \cdot A_3 + S_0' \cdot S_1' \cdot S_2 \cdot A_4 + S_0 \cdot S_1' \cdot S_2 \cdot A_5 + S_0' \cdot S_1 \cdot S_2 \cdot A_6 + S_0 \cdot S_1 \cdot S_2 \cdot A_7$$

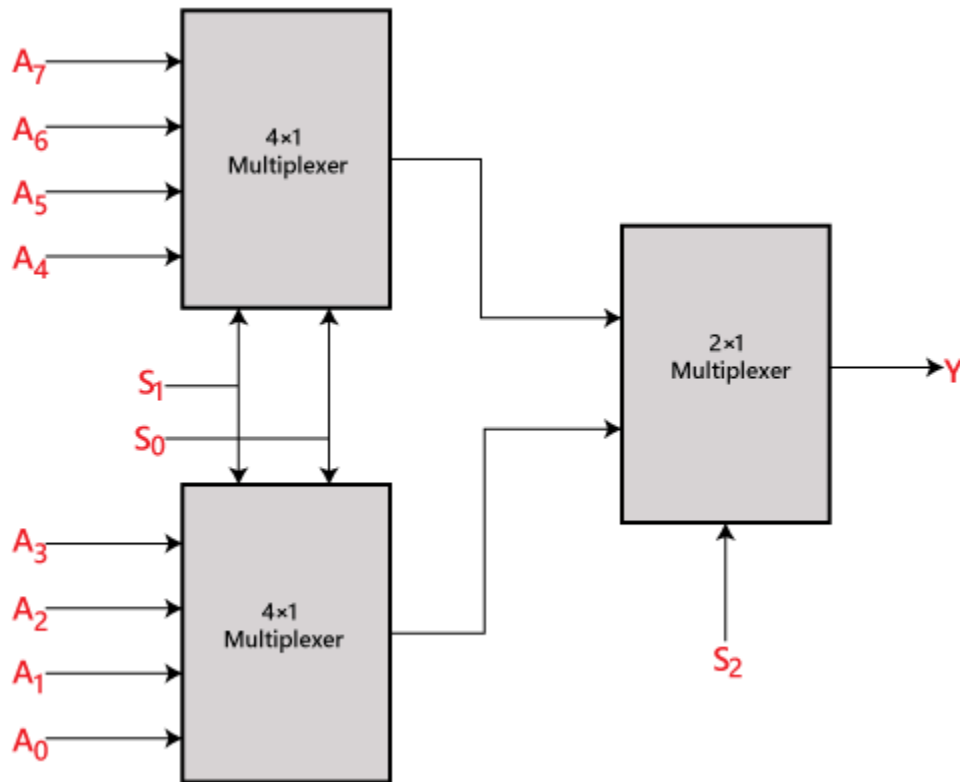
Logical circuit of the above expression is given below:



## 8 ×1 multiplexer using 4×1 and 2×1 multiplexer

We can implement the 8×1 multiplexer using a lower order multiplexer. To implement the 8×1 multiplexer, we need two 4×1 multiplexers and one 2×1 multiplexer. The 4×1 multiplexer has 2 selection lines, 4 inputs, and 1 output. The 2×1 multiplexer has only 1 selection line.

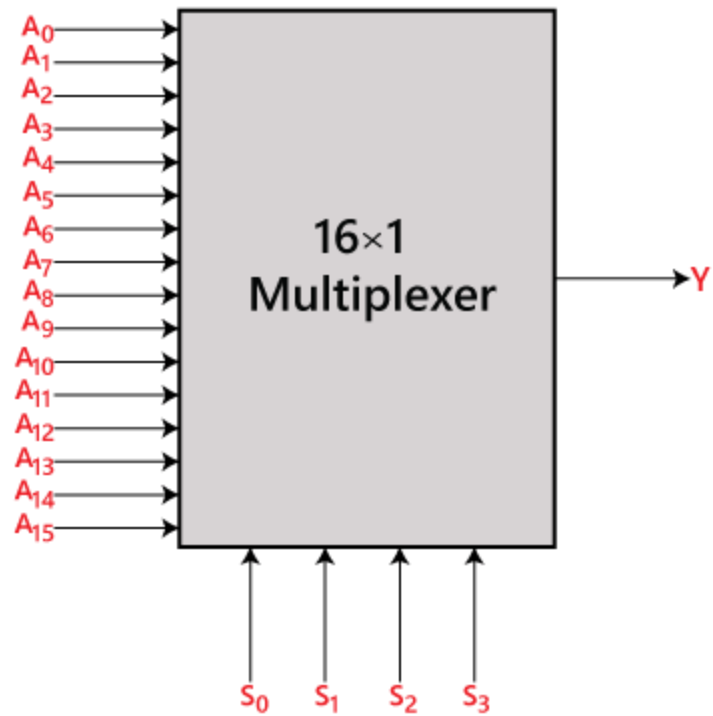
For getting 8 data inputs, we need two  $4 \times 1$  multiplexers. The  $4 \times 1$  multiplexer produces one output. So, in order to get the final output, we need a  $2 \times 1$  multiplexer. The block diagram of  $8 \times 1$  multiplexer using  $4 \times 1$  and  $2 \times 1$  multiplexer is given below.



## 16 to 1 Multiplexer

In the 16 to 1 multiplexer, there are total of 16 inputs, i.e.,  $A_0, A_1, \dots, A_{15}$ , 4 selection lines, i.e.,  $S_0, S_1, S_2$ , and  $S_3$  and single output, i.e.,  $Y$ . On the basis of the combination of inputs that are present at the selection lines  $S_0, S_1$ , and  $S_2$ , one of these 16 inputs will be connected to the output. The block diagram and the truth table of the  $16 \times 1$

**Block Diagram:**



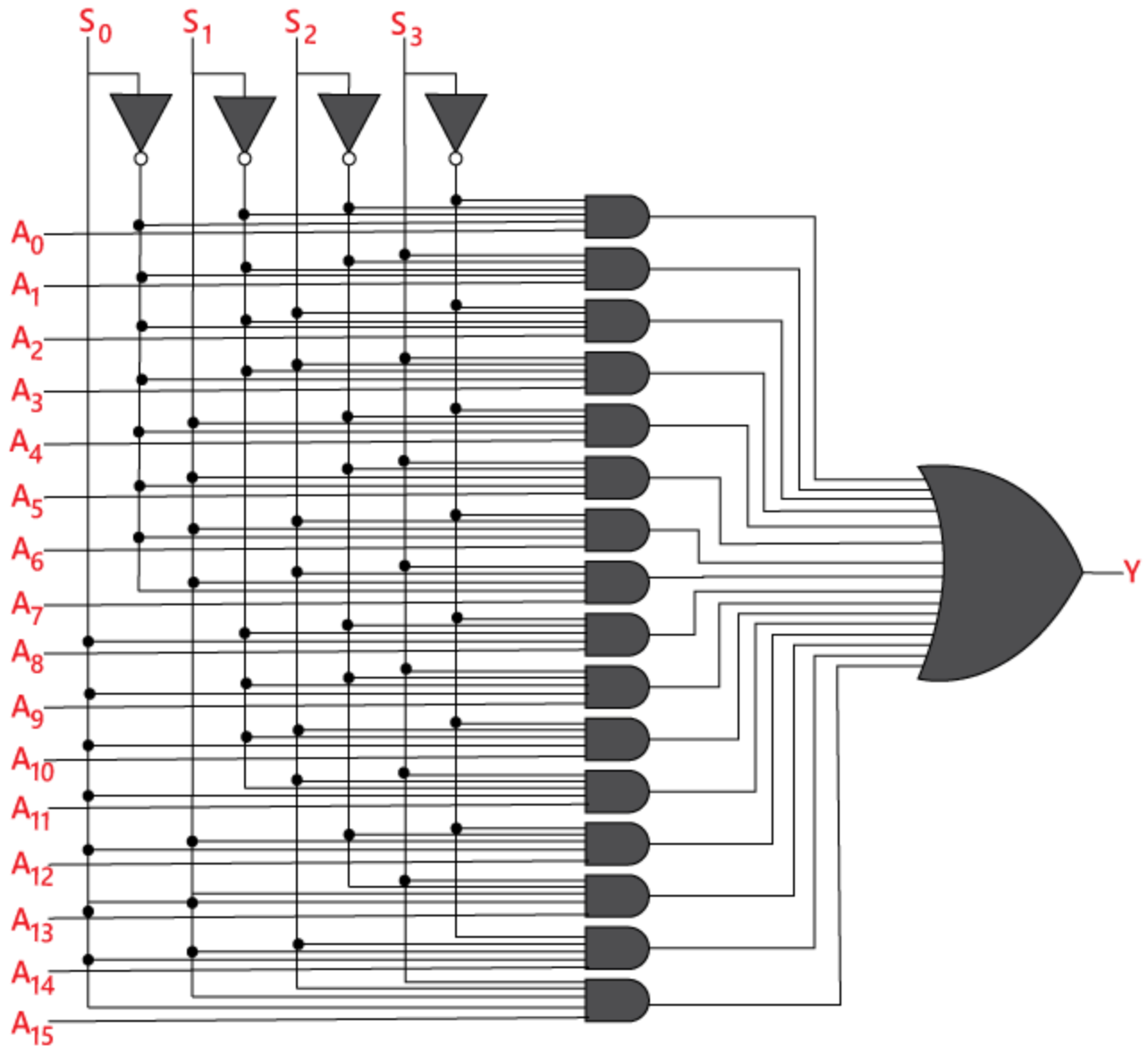
Truth Table:

| INPUTS         |                |                |                | Output          |
|----------------|----------------|----------------|----------------|-----------------|
| S <sub>0</sub> | S <sub>1</sub> | S <sub>2</sub> | S <sub>3</sub> | Y               |
| 0              | 0              | 0              | 0              | A <sub>0</sub>  |
| 0              | 0              | 0              | 1              | A <sub>1</sub>  |
| 0              | 0              | 1              | 0              | A <sub>2</sub>  |
| 0              | 0              | 1              | 1              | A <sub>3</sub>  |
| 0              | 1              | 0              | 0              | A <sub>4</sub>  |
| 0              | 1              | 0              | 1              | A <sub>5</sub>  |
| 0              | 1              | 1              | 0              | A <sub>6</sub>  |
| 0              | 1              | 1              | 1              | A <sub>7</sub>  |
| 1              | 0              | 0              | 0              | A <sub>8</sub>  |
| 1              | 0              | 0              | 1              | A <sub>9</sub>  |
| 1              | 0              | 1              | 0              | A <sub>10</sub> |
| 1              | 0              | 1              | 1              | A <sub>11</sub> |
| 1              | 1              | 0              | 0              | A <sub>12</sub> |
| 1              | 1              | 0              | 1              | A <sub>13</sub> |
| 1              | 1              | 1              | 0              | A <sub>14</sub> |
| 1              | 1              | 1              | 1              | A <sub>15</sub> |

The logical expression of the term Y is as follows:

$$Y = A_0.S_0'.S_1'.S_2'.S_3' + A_1.S_0'.S_1'.S_2'.S_3 + A_2.S_0'.S_1'.S_2.S_3' + A_3.S_0'.S_1'.S_2.S_3 + A_4.S_0'.S_1.S_2'.S_3' + A_5.S_0'.S_1.S_2'.S_3 + A_6.S_1.S_2.S_3' + A_7.S_0'.S_1.S_2.S_3 + A_8.S_0.S_1'.S_2'.S_3' + A_9.S_0.S_1'.S_2'.S_3 + A_{10}.S_0.S_1'.S_2.S_3' + A_{11}.S_0.S_1'.S_2.S_3 + A_{12}.S_0.S_1.S_2'.S_3' + A_{13}.S_0.S_1.S_2'.S_3 + A_{14}.S_0.S_1.S_2.S_3' + A_{15}.S_0.S_1.S_2'.S_3$$

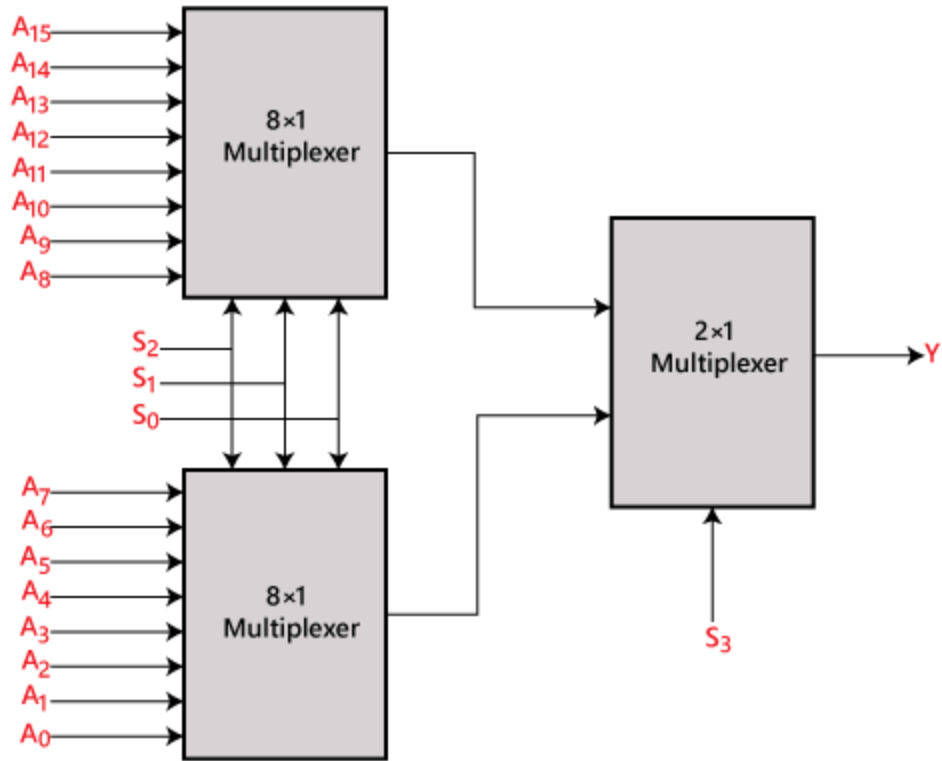
Logical circuit of the above expression is given below:



### 16×1 multiplexer using 8×1 and 2×1 multiplexer

We can implement the 16×1 multiplexer using a lower order multiplexer. To implement the 8×1 multiplexer, we need two 8×1 multiplexers and one 2×1 multiplexer. The 8×1 multiplexer has 3 selection lines, 4 inputs, and 1 output. The 2×1 multiplexer has only 1 selection line.

For getting 16 data inputs, we need two 8 ×1 multiplexers. The 8×1 multiplexer produces one output. So, in order to get the final output, we need a 2×1 multiplexer. The block diagram of 16×1 multiplexer using 8×1 and 2×1 multiplexer is given below.



# De-multiplexer

A De-multiplexer is a combinational circuit that has only 1 input line and  $2^N$  output lines. Simply, the multiplexer is a single-input and multi-output combinational circuit. The information is received from the single input lines and directed to the output line. On the basis of the values of the selection lines, the input will be connected to one of these outputs. De-multiplexer is opposite to the multiplexer.

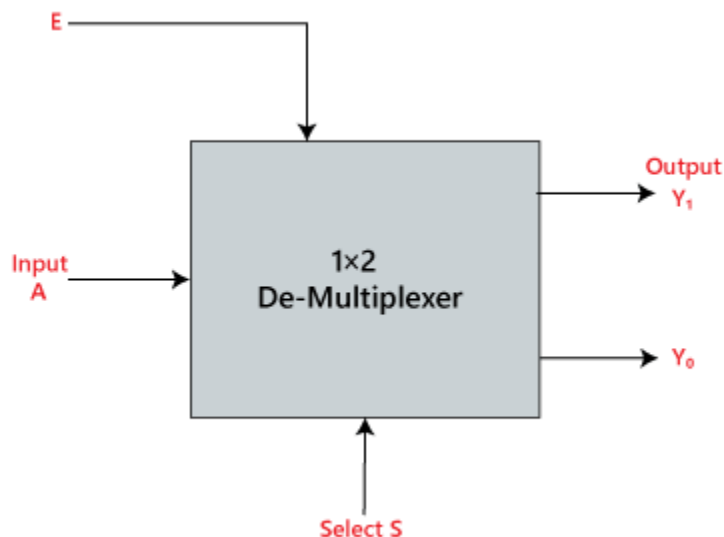
Unlike encoder and decoder, there are  $n$  selection lines and  $2^n$  outputs. So, there is a total of  $2^n$  possible combinations of inputs. De-multiplexer is also treated as **De-mux**.

There are various types of De-multiplexer which are as follows:

## 1×2 De-multiplexer:

In the 1 to 2 De-multiplexer, there are only two outputs, i.e.,  $Y_0$ , and  $Y_1$ , 1 selection lines, i.e.,  $S_0$ , and single input, i.e.,  $A$ . On the basis of the selection value, the input will be connected to one of the outputs. The block diagram and the truth table of the 1×2 multiplexer are given below.

### Block Diagram:



### Truth Table:

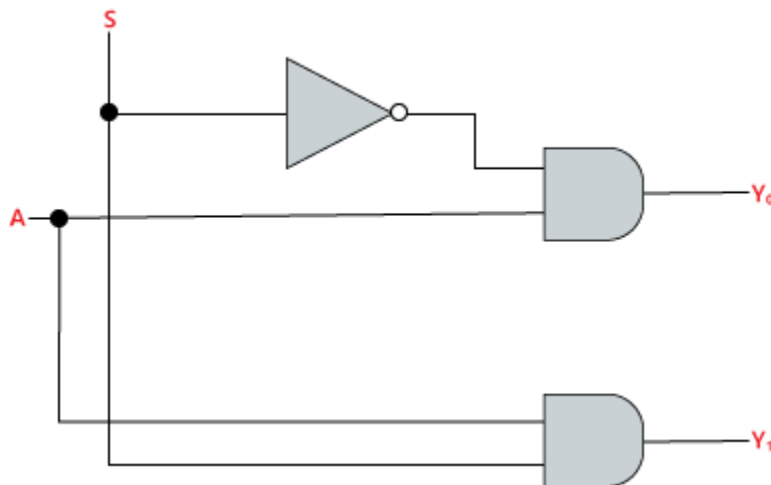
| INPUTS | Output |       |
|--------|--------|-------|
| $S_0$  | $Y_1$  | $Y_0$ |
| 0      | 0      | A     |
| 1      | A      | 0     |

The logical expression of the term Y is as follows:

$$Y_0 = S_0' \cdot A$$

$$Y_1 = S_0 \cdot A$$

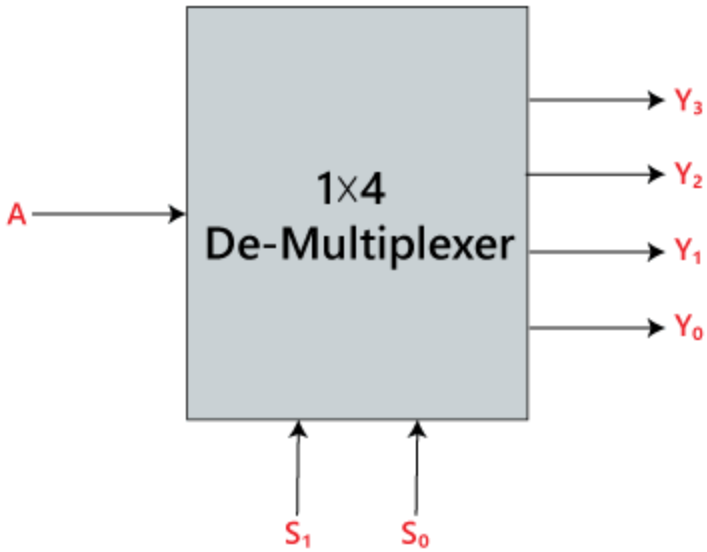
Logical circuit of the above expressions is given below:



## 1×4 De-multiplexer:

In 1 to 4 De-multiplexer, there are total of four outputs, i.e.,  $Y_0$ ,  $Y_1$ ,  $Y_2$ , and  $Y_3$ , 2 selection lines, i.e.,  $S_0$  and  $S_1$  and single input, i.e., A. On the basis of the combination of inputs which are present at the selection lines  $S_0$  and  $S_1$ , the input be connected to one of the outputs. The block diagram and the truth table of the 1×4 multiplexer are given below.

**Block Diagram:**



Truth Table:

| INPUTS         |                | Output         |                |                |                |
|----------------|----------------|----------------|----------------|----------------|----------------|
| S <sub>1</sub> | S <sub>0</sub> | Y <sub>3</sub> | Y <sub>2</sub> | Y <sub>1</sub> | Y <sub>0</sub> |
| 0              | 0              | 0              | 0              | 0              | A              |
| 0              | 1              | 0              | 0              | A              | 0              |
| 1              | 0              | 0              | A              | 0              | 0              |
| 1              | 1              | A              | 0              | 0              | 0              |

The logical expression of the term Y is as follows:

$$Y_0 = S_1'$$

$$S_0'$$

$$A$$

$$y_1 = S_1'$$

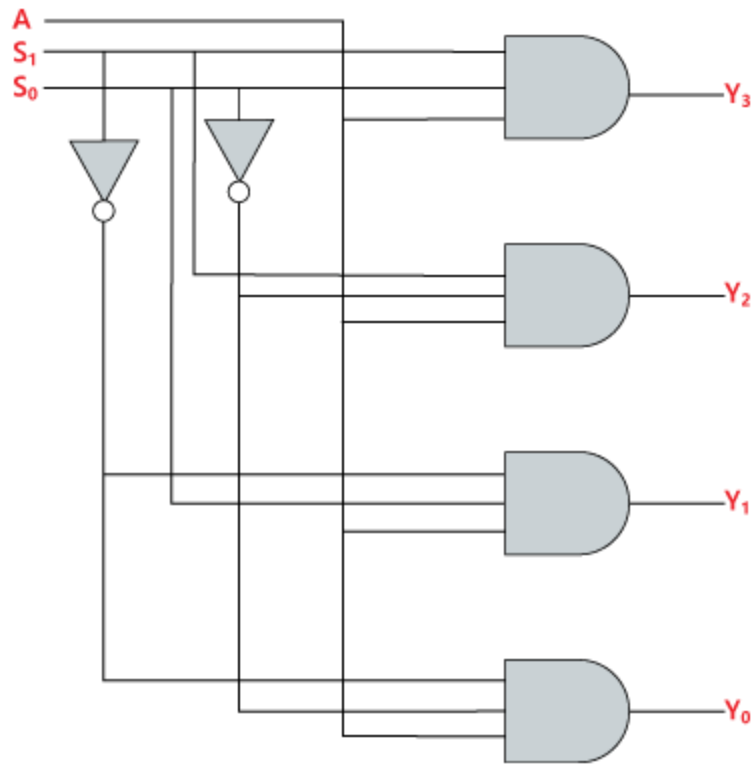
$$S_0 A$$

$$y_2 = S_1 S_0'$$

$$A$$

$$y_3 = S_1 S_0 A$$

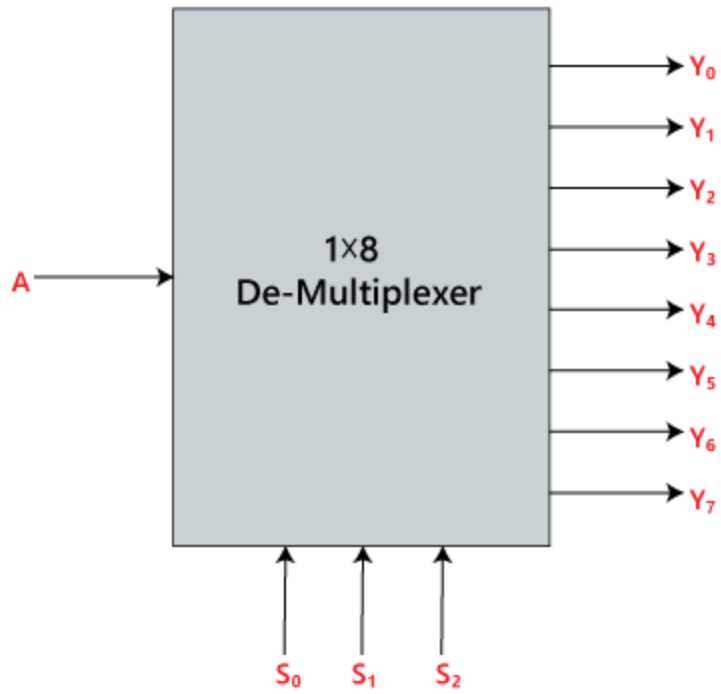
Logical circuit of the above expressions is given below:



## 1×8 De-multiplexer

In 1 to 8 De-multiplexer, there are total of eight outputs, i.e.,  $Y_0$ ,  $Y_1$ ,  $Y_2$ ,  $Y_3$ ,  $Y_4$ ,  $Y_5$ ,  $Y_6$ , and  $Y_7$ , 3 selection lines, i.e.,  $S_0$ ,  $S_1$  and  $S_2$  and single input, i.e.,  $A$ . On the basis of the combination of inputs which are present at the selection lines  $S^0$ ,  $S^1$  and  $S_2$ , the input will be connected to one of these outputs. The block diagram and the truth table of the 1×8 de-multiplexer are given below.

**Block Diagram:**



Truth Table:

| INPUTS |       |       | Output |       |       |       |       |       |       |       |
|--------|-------|-------|--------|-------|-------|-------|-------|-------|-------|-------|
| $S_2$  | $S_1$ | $S_0$ | $Y_7$  | $Y_6$ | $Y_5$ | $Y_4$ | $Y_3$ | $Y_2$ | $Y_1$ | $Y_0$ |
| 0      | 0     | 0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | $A$   |
| 0      | 0     | 1     | 0      | 0     | 0     | 0     | 0     | 0     | $A$   | 0     |
| 0      | 1     | 0     | 0      | 0     | 0     | 0     | 0     | $A$   | 0     | 0     |
| 0      | 1     | 1     | 0      | 0     | 0     | 0     | $A$   | 0     | 0     | 0     |
| 1      | 0     | 0     | 0      | 0     | 0     | $A$   | 0     | 0     | 0     | 0     |
| 1      | 0     | 1     | 0      | 0     | $A$   | 0     | 0     | 0     | 0     | 0     |
| 1      | 1     | 0     | 0      | $A$   | 0     | 0     | 0     | 0     | 0     | 0     |
| 1      | 1     | 1     | $A$    | 0     | 0     | 0     | 0     | 0     | 0     | 0     |

The logical expression of the term  $Y$  is as follows:

$$Y_0 = S_0' \cdot S_1' \cdot S_2' \cdot A$$

$$Y_1 = S_0 \cdot S_1' \cdot S_2' \cdot A$$

$$Y_2 = S_0' \cdot S_1 \cdot S_2' \cdot A$$

$$Y_3 = S_0 \cdot S_1 \cdot S_2' \cdot A$$

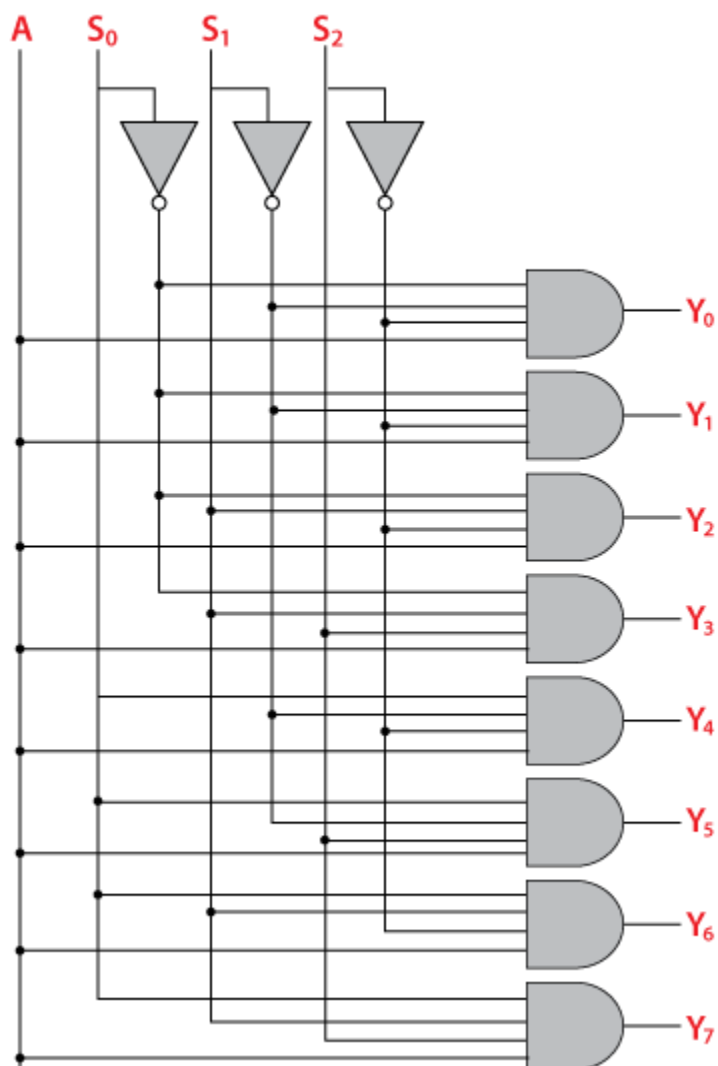
$$Y_4 = S_0' \cdot S_1' \cdot S_2 \cdot A$$

$$Y_5 = S_0 \cdot S_1' \cdot S_2 \cdot A$$

$$Y_6 = S_0' \cdot S_1 \cdot S_2 \cdot A$$

$$Y_7 = S_0 \cdot S_1 \cdot S_2 \cdot A$$

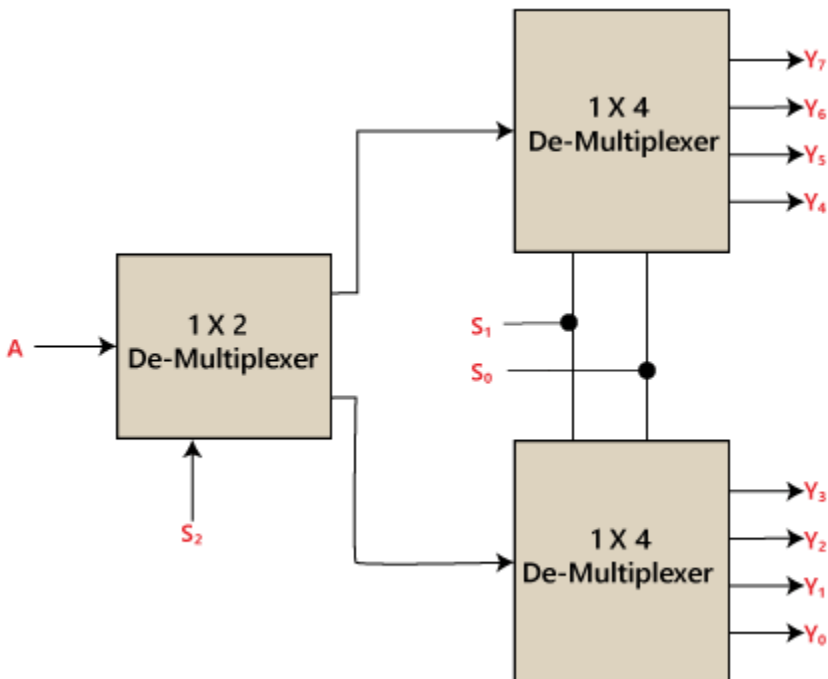
Logical circuit of the above expressions is given below:



1x8 De-multiplexer using 1x4 and 1x2 de-multiplexer

We can implement the  $1 \times 8$  de-multiplexer using a lower order de-multiplexer. To implement the  $1 \times 8$  de-multiplexer, we need two  $1 \times 4$  de-multiplexer and one  $1 \times 2$  de-multiplexer. The  $1 \times 4$  multiplexer has 2 selection lines, 4 outputs, and 1 input. The  $1 \times 2$  de-multiplexer has only 1 selection line.

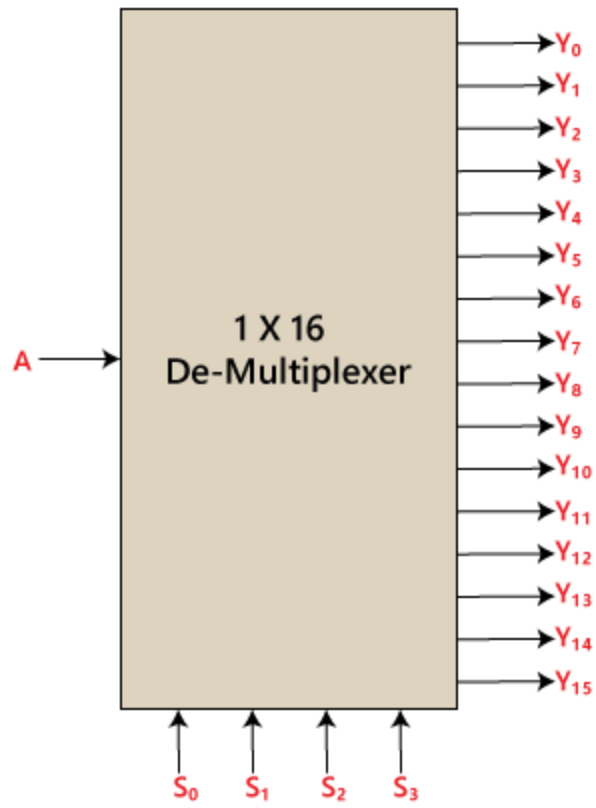
For getting 8 data outputs, we need two  $1 \times 4$  de-multiplexer. The  $1 \times 2$  de-multiplexer produces two outputs. So, in order to get the final output, we have to pass the outputs of  $1 \times 2$  de-multiplexer as an input of both the  $1 \times 4$  de-multiplexer. The block diagram of  $1 \times 8$  de-multiplexer using  $1 \times 4$  and  $1 \times 2$  de-multiplexer is given below.



## 1 x 16 De-multiplexer

In  $1 \times 16$  de-multiplexer, there are total of 16 outputs, i.e.,  $Y_0, Y_1, \dots, Y_{16}$ , 4 selection lines, i.e.,  $S_0, S_1, S_2$ , and  $S_3$  and single input, i.e.,  $A$ . On the basis of the combination of inputs which are present at the selection lines  $S^0, S^1$ , and  $S_2$ , the input will be connected to one of these outputs. The block diagram and the truth table of the  $1 \times 16$  de-multiplexer are given below.

**Block Diagram:**



Truth Table:

| INPUTS         |                |                |                | OUTPUTS         |                 |                 |                 |                 |                 |                |                |                |                |                |                |                |                |                |                |
|----------------|----------------|----------------|----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| S <sub>3</sub> | S <sub>2</sub> | S <sub>1</sub> | S <sub>0</sub> | Y <sub>15</sub> | Y <sub>14</sub> | Y <sub>13</sub> | Y <sub>12</sub> | Y <sub>11</sub> | Y <sub>10</sub> | Y <sub>9</sub> | Y <sub>8</sub> | Y <sub>7</sub> | Y <sub>6</sub> | Y <sub>5</sub> | Y <sub>4</sub> | Y <sub>3</sub> | Y <sub>2</sub> | Y <sub>1</sub> | Y <sub>0</sub> |
| 0              | 0              | 0              | 0              | 0               | 0               | 0               | 0               | 0               | 0               | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | A              |
| 0              | 0              | 0              | 1              | 0               | 0               | 0               | 0               | 0               | 0               | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | A              | 0              |
| 0              | 0              | 1              | 0              | 0               | 0               | 0               | 0               | 0               | 0               | 0              | 0              | 0              | 0              | 0              | 0              | 0              | A              | 0              | 0              |
| 0              | 0              | 1              | 1              | 0               | 0               | 0               | 0               | 0               | 0               | 0              | 0              | 0              | 0              | 0              | 0              | A              | 0              | 0              | 0              |
| 0              | 1              | 0              | 0              | 0               | 0               | 0               | 0               | 0               | 0               | 0              | 0              | 0              | 0              | 0              | A              | 0              | 0              | 0              | 0              |
| 0              | 1              | 0              | 1              | 0               | 0               | 0               | 0               | 0               | 0               | 0              | 0              | 0              | 0              | A              | 0              | 0              | 0              | 0              | 0              |
| 0              | 1              | 1              | 0              | 0               | 0               | 0               | 0               | 0               | 0               | 0              | 0              | 0              | A              | 0              | 0              | 0              | 0              | 0              | 0              |
| 0              | 1              | 1              | 1              | 0               | 0               | 0               | 0               | 0               | 0               | 0              | 0              | A              | 0              | 0              | 0              | 0              | 0              | 0              | 0              |
| 1              | 0              | 0              | 0              | 0               | 0               | 0               | 0               | 0               | 0               | 0              | A              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              |
| 1              | 0              | 0              | 1              | 0               | 0               | 0               | 0               | 0               | 0               | A              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              |
| 1              | 0              | 1              | 0              | 0               | 0               | 0               | 0               | 0               | A               | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              |
| 1              | 0              | 1              | 1              | 0               | 0               | 0               | 0               | A               | 0               | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              |
| 1              | 1              | 0              | 0              | 0               | 0               | 0               | A               | 0               | 0               | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              |
| 1              | 1              | 0              | 1              | 0               | 0               | A               | 0               | 0               | 0               | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              |
| 1              | 1              | 1              | 0              | 0               | A               | 0               | 0               | 0               | 0               | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              |
| 1              | 1              | 1              | 1              | A               | 0               | 0               | 0               | 0               | 0               | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              |

The logical expression of the term Y is as follows:

$$Y_0 = A.S_0'.S_1'.S_2'.S_3'$$

$$Y_1 = A.S_0'.S_1'.S_2'.S_3$$

$$Y_2 = A.S_0'.S_1'.S_2.S_3'$$

$$Y_3 = A.S_0'.S_1'.S_2.S_3$$

$$Y_4 = A.S_0'.S_1.S_2'.S_3'$$

$$Y_5 = A.S_0'.S_1.S_2'.S_3$$

$$Y_6 = A.S_0'.S_1.S_2.S_3'$$

$$Y_7 = A.S_0'.S_1.S_2.S_3$$

$$Y_8 = A.S_0.S_1'.S_2'.S_3'$$

$$Y_9 = A.S_0.S_1'.S_2'.S_3$$

$$Y_{10} = A.S_0.S_1'.S_2.S_3'$$

$$Y_{11} = A.S_0.S_1'.S_2'.S_3$$

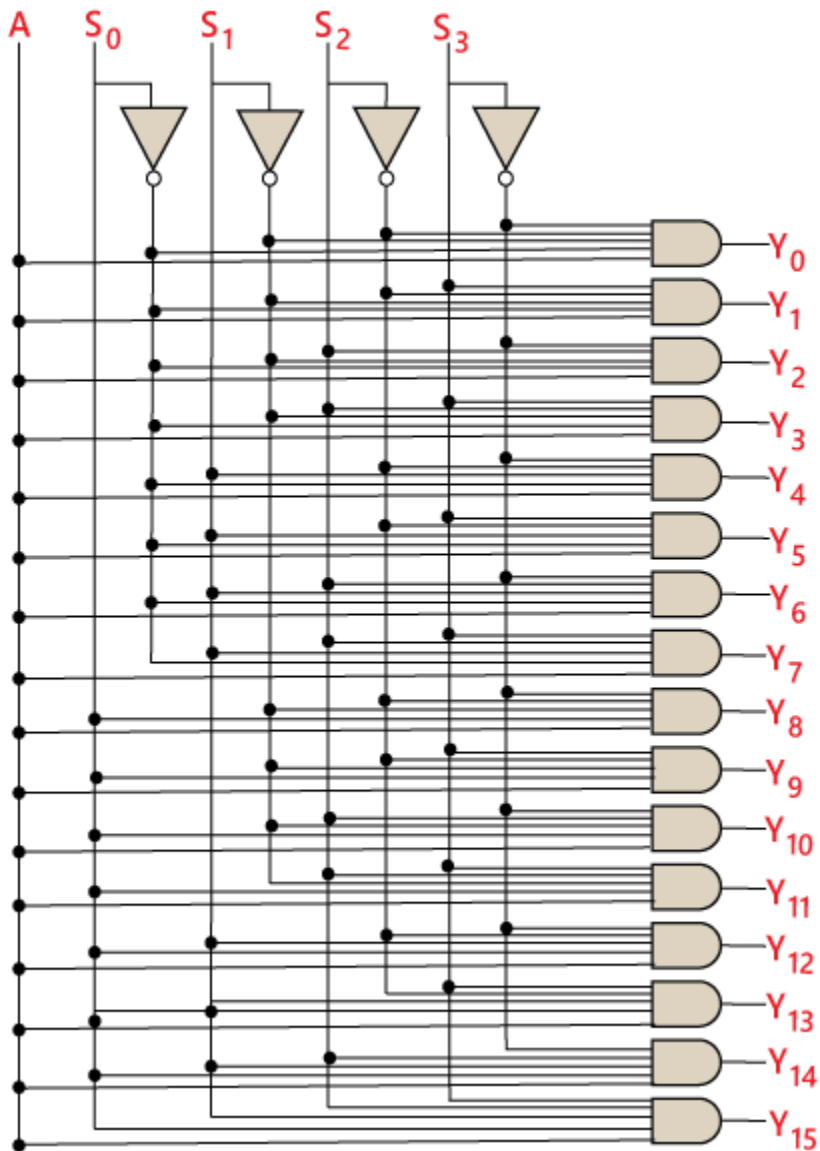
$$Y_{12} = A.S_0.S_1.S_2'.S_3'$$

$$Y_{13} = A.S_0.S_1.S_2'.S_3$$

$$Y_{14} = A.S_0.S_1.S_2.S_3'$$

$$Y_{15} = A.S_0.S_1.S_2'.S_3$$

Logical circuit of the above expressions is given below:



1x16 de-multiplexer using 1x8 and 1x2 de-multiplexer

We can implement the  $1 \times 16$  de-multiplexer using a lower order de-multiplexer. To implement the  $1 \times 16$  de-multiplexer, we need two  $1 \times 8$  de-multiplexer and one  $1 \times 2$  de-multiplexer. The  $1 \times 8$  multiplexer has 3 selection lines, 1 input, and 8 outputs. The  $1 \times 2$  de-multiplexer has only 1 selection line.

For getting 16 data outputs, we need two  $1 \times 8$  de-multiplexer. The  $1 \times 8$  de-multiplexer produces eight outputs. So, in order to get the final output, we need a  $1 \times 2$  de-multiplexer to produce two outputs from a single input. Then we pass these outputs into both the de-multiplexer as an input. The block diagram of  $1 \times 16$  de-multiplexer using  $1 \times 8$  and  $1 \times 2$  de-multiplexer is given below.

